Question	Answer	Marks
4(a)	(defining equation of s.h.m. is) $a = -kx$ where k is a constant **or** $a \propto -x$	**B1**
	g and L are constant (so $a \propto -x$ and hence s.h.m.)	**B1**
4(b)	$T = 0.50$ s **and** $T = 2\pi / \omega$	**C1**
	$\omega^2 = 2g / L$	**C1**
	$L = (2 \times 9.81 \times 0.50^2) / 4\pi^2$ $= 0.12$ m	**A1**
4(c)(i)	Any one from: • viscosity of liquid • friction within the liquid • viscous drag • friction/resistance between walls of tube and liquid	**B1**
4(c)(ii)	(maximum) KE = $\frac{1}{2}mv_0^2$ **and** $v_0 = \omega x_0$ **or** energy = $\frac{1}{2}m\omega^2 x_0^2$	**C1**
	ratio = $(1.3 / 2.0)^2$ $= 0.42$	**A1**

Question	Answer	Marks
5(a)	amplitude of carrier (wave) varies	B1